

Lecture 9: Review of surface geometry

Let $f : U \rightarrow M \subset \mathbb{R}^3$ be a regular surface element. Write $q = (u, v) \in U$ and $p = f(q)$. A **unit normal** at p is a vector $\nu(q)$ satisfying

$$\|\nu(q)\| = 1, \quad \langle \nu(q), X \rangle = 0 \quad \text{for all } X \in T_p M.$$

There are two choices at each point. Following Kühnel, we view the normal as a map $\nu : U \rightarrow S^2$. One smooth choice is

$$\nu(u, v) = \frac{f_u \times f_v}{\|f_u \times f_v\|}.$$

The denominator is nonzero because f_u, f_v are linearly independent. Reversing the order of the parameters reverses this normal.

For the graph $f(u, v) = (u, v, h(u, v))$, the upward normal is

$$\nu = \frac{(-h_u, -h_v, 1)}{\sqrt{1 + h_u^2 + h_v^2}}.$$

For a regular level surface $M = F^{-1}(0)$, we can instead use

$$\nu(q) = \frac{\nabla F(f(q))}{\|\nabla F(f(q))\|}.$$

A global smooth choice of unit normal exists exactly when the surface is orientable. Local choices need not fit together globally.

The *Möbius strip* illustrates the obstruction to choosing a global unit normal. It is the image of

$$f : \mathbb{R} \times (-\varepsilon, \varepsilon) \longrightarrow \mathbb{R}^3, \quad f(u, v) = \begin{pmatrix} (1 + v \sin(u/2)) \sin u \\ (1 + v \sin(u/2)) \cos u \\ v \cos(u/2) \end{pmatrix},$$

for sufficiently small $\varepsilon > 0$. This map is regular, but is not globally injective: after one circuit,

$$f(u + 2\pi, v) = f(u, -v).$$

It suffices to study the normal along the central circle $v = 0$. There the coordinate tangent vectors are

$$f_u(u, 0) = (\cos u, -\sin u, 0),$$

$$f_v(u, 0) = (\sin(u/2) \sin u, \sin(u/2) \cos u, \cos(u/2)).$$

These vectors are orthogonal and have unit length, so their cross product is already a unit normal:

$$\nu(u, 0) = f_u(u, 0) \times f_v(u, 0) = \begin{pmatrix} -\cos(u/2) \sin u \\ -\cos(u/2) \cos u \\ \sin(u/2) \end{pmatrix}.$$

This normal varies continuously as u runs from 0 to 2π .

At the two ends of the parameter interval, we have

$$f(0, 0) = f(2\pi, 0) = (0, 1, 0),$$

but

$$\nu(0, 0) = (0, -1, 0), \quad \nu(2\pi, 0) = (0, 1, 0) = -\nu(0, 0).$$

Thus the normal on the parameter domain does not define a single-valued normal on the image.

To rule out *every* choice, suppose a continuous unit normal field N existed on the Möbius strip. At each point of the central circle there are exactly two unit normals, so

$$N(f(u, 0)) = \sigma(u)\nu(u, 0), \quad \sigma(u) \in \{1, -1\}.$$

The function

$$\sigma(u) = \langle N(f(u, 0)), \nu(u, 0) \rangle$$

is continuous. Since $[0, 2\pi]$ is connected, σ must be constant. Consequently,

$$N(f(2\pi, 0)) = -N(f(0, 0)).$$

But the arguments are the same surface point. This would force a unit vector to equal its negative, which is impossible.

There is no continuous unit normal field on the Möbius strip, so it is ***non-orientable***. The obstruction is global: every sufficiently small surface patch still has a unit normal.

For the surface element $f : U \rightarrow M$ and its normal map $\nu : U \rightarrow S^2$, the **Weingarten map**, or **shape operator**, is

$$L_q = -D\nu_q \circ (Df_q)^{-1} : T_pM \longrightarrow T_pM, \quad p = f(q).$$

The inverse means the inverse of the linear isomorphism

$$Df_q : \mathbb{R}^2 \longrightarrow T_pM.$$

It is defined on the tangent plane, not on all of $\mathbb{R}^3$. For $X = Df_q(a)$, the definition says

$$L_q(X) = -D\nu_q(a).$$

First recover the parameter velocity a ; then measure the corresponding change in the normal, with a minus sign.

Differentiating $\langle \nu, \nu \rangle = 1$ gives $\langle D\nu_q(a), \nu(q) \rangle = 0$. Thus $D\nu_q(a)$ lies in $\nu(q)^\perp = T_pM$, as required.

In particular, in the coordinate tangent basis,

$$L(f_u) = -\nu_u, \quad L(f_v) = -\nu_v.$$

If N denotes the normal field on the surface itself, then $\nu = N \circ f$. The chain rule gives $L_q = -dN_p$, which is the same operator written without parameters.

Consider the sphere of radius $R > 0$, centered at the origin:

$$M = \{p \in \mathbb{R}^3 : \|p\|^2 = R^2\}.$$

Let $f : U \rightarrow M$ be any regular surface element. Differentiating $\langle f, f \rangle = R^2$ gives

$$\langle f, f_u \rangle = \langle f, f_v \rangle = 0.$$

Thus f is perpendicular to the tangent plane. Since $\|f\| = R$, the outward unit normal is

$$\nu = \frac{f}{R}, \quad D\nu_q = \frac{1}{R}Df_q.$$

Substitute this into the definition of L :

$$\begin{aligned} L_q &= -D\nu_q \circ (Df_q)^{-1} \\ &= -\frac{1}{R}Df_q \circ (Df_q)^{-1} \\ &= -\frac{1}{R} \text{id}_{T_{f(q)}M}. \end{aligned}$$

In other words, $L_q(X) = -X/R$ for every tangent vector X . The normal changes at the same rate in every tangent direction.

For the inward normal $\nu = -f/R$, the same calculation gives $L = R^{-1} \text{id}$. The sign depends on the normal choice; the magnitude $1/R$ does not.

This computation works in any regular parametrization: we never need to write down spherical coordinates or invert a matrix.

For a plane, take a parametrization

$$f(u, v) = p_0 + u a + v b,$$

where $a, b \in \mathbb{R}^3$ are fixed, linearly independent vectors. Then

$$f_u = a, \quad f_v = b, \quad \nu = \frac{a \times b}{\|a \times b\|}.$$

The normal is constant, so its differential vanishes:

$$\nu_u = \nu_v = 0, \quad D\nu = 0.$$

Consequently,

$$L = -D\nu \circ (Df)^{-1} = 0.$$

Equivalently, the coordinate-basis formula gives

$$L(a) = L(f_u) = -\nu_u = 0, \quad L(b) = L(f_v) = -\nu_v = 0.$$

Since a, b span the tangent plane, $L(X) = 0$ for every tangent vector X .

Geometrically, moving in any direction within the plane leaves the normal unchanged. Reversing the normal still gives $L = 0$. Both eigenvalues are zero.

For the circular cylinder $x^2 + y^2 = R^2$, with $R > 0$, use

$$f(u, v) = (R \cos u, R \sin u, v).$$

Restrict u to an angular interval to obtain a local chart. Its coordinate tangent vectors and outward unit normal are

$$\begin{aligned} f_u &= (-R \sin u, R \cos u, 0), \\ f_v &= (0, 0, 1), \quad \nu = (\cos u, \sin u, 0). \end{aligned}$$

Differentiate the normal:

$$\nu_u = (-\sin u, \cos u, 0) = \frac{1}{R} f_u, \quad \nu_v = 0.$$

We can compute L on a basis without explicitly finding $(Df)^{-1}$:

$$L(f_u) = -\nu_u = -\frac{1}{R} f_u, \quad L(f_v) = -\nu_v = 0.$$

Thus, in the orthonormal basis $e_1 = f_u/R$, $e_2 = f_v$,

$$[L] = \begin{pmatrix} -1/R & 0 \\ 0 & 0 \end{pmatrix}.$$

Moving around a horizontal circle changes the normal; moving parallel to the axis does not. The eigenvalues are $-1/R$ and 0 . With the inward normal they become $1/R$ and 0 .

The first fundamental form on tangent vectors is

$$I(X, Y) = \langle X, Y \rangle.$$

The Weingarten map is ***self-adjoint*** with respect to this inner product:

$$I(LX, Y) = I(X, LY) \quad \text{for all } X, Y \in T_p M.$$

Proof. Write $u_1 = u, u_2 = v, f_i = \partial f / \partial u_i$, and $v_i = \partial v / \partial u_i$. Since $Lf_i = -v_i$,

$$I(Lf_i, f_j) = -\langle v_i, f_j \rangle.$$

Differentiate the identity $\langle v, f_j \rangle = 0$ in the u_i -direction:

$$\langle v_i, f_j \rangle + \langle v, f_{ij} \rangle = 0.$$

The mixed partial derivatives commute, so

$$\begin{aligned} I(Lf_i, f_j) &= \langle v, f_{ij} \rangle \\ &= \langle v, f_{ji} \rangle \\ &= -\langle f_i, v_j \rangle = I(f_i, Lf_j). \end{aligned}$$

The vectors f_1, f_2 form a basis. Bilinearity therefore extends this equality to all tangent vectors X, Y .

Consequently L has real eigenvalues and an orthonormal eigenbasis for I . Its eigenvalues κ_1, κ_2 are the ***principal curvatures***.

The *second fundamental form* is

$$\Pi(X, Y) = I(LX, Y).$$

Self-adjointness of L gives its symmetry:

$$\Pi(X, Y) = I(X, LY) = \Pi(Y, X).$$

Its coefficients in the basis f_1, f_2 are

$$h_{ij} = \Pi(f_i, f_j) = \langle \nu, f_{ij} \rangle = -\langle \nu_i, f_j \rangle.$$

Write $g = (g_{ij})$, with $g_{ij} = \langle f_i, f_j \rangle$, and $h = (h_{ij})$. If A is the matrix of L in this basis, then

$$h = A^\top g = gA, \quad A = g^{-1}h.$$

Thus A need not be an ordinary symmetric matrix: the basis f_1, f_2 need not be orthonormal. Self-adjointness means $A^\top g = gA$, rather than $A^\top = A$.

For a unit tangent vector T , the *normal curvature* is

$$\kappa_\nu(T) = \Pi(T, T).$$

It measures bending in the normal direction, rather than the full curvature of a curve in space.

The *third fundamental form* is

$$\text{III}(X, Y) = \text{I}(LX, LY).$$

By self-adjointness, this can also be written as

$$\text{III}(X, Y) = \text{I}(L^2X, Y).$$

It is a symmetric bilinear form, with

$$\text{III}(X, X) = \|LX\|^2 \geq 0.$$

It is positive definite precisely when L is invertible. Otherwise it is degenerate; on a plane it vanishes identically.

In the coordinate tangent basis, $Lf_i = -v_i$, so

$$e_{ij} = \text{III}(f_i, f_j) = \langle v_i, v_j \rangle.$$

Equivalently, its matrix is

$$e = A^\top g A = h g^{-1} h.$$

Thus the third fundamental form measures lengths and angles of the *changes in the normal*. In coordinates it is the pullback of the round metric on S^2 by the Gauss map $\nu : U \rightarrow S^2$; the Gauss map need not be an immersion.

For a sphere of radius R , $L = \pm R^{-1} \text{id}$, so $\text{III} = R^{-2}\text{I}$. Reversing the normal reverses II , but leaves I and III unchanged.

The *mean curvature* and *Gaussian curvature* are

$$H = \frac{\kappa_1 + \kappa_2}{2} = \frac{1}{2} \operatorname{tr} L, \quad K = \kappa_1 \kappa_2 = \det L.$$

If e_1, e_2 is an orthonormal principal basis, then

$$\kappa_\nu(\cos \theta e_1 + \sin \theta e_2) = \kappa_1 \cos^2 \theta + \kappa_2 \sin^2 \theta.$$

Averaging this expression over $0 \leq \theta \leq 2\pi$ gives H . This explains the word “mean.”

For computations, if

$$g = \begin{pmatrix} E & F \\ F & G \end{pmatrix}, \quad h = \begin{pmatrix} h_{11} & h_{12} \\ h_{12} & h_{22} \end{pmatrix},$$

then

$$H = \frac{Gh_{11} - 2Fh_{12} + Eh_{22}}{2(EG - F^2)}, \quad K = \frac{h_{11}h_{22} - h_{12}^2}{EG - F^2}.$$

Reversing ν reverses L, κ_1, κ_2, H , but leaves K unchanged. In particular, the condition $H = 0$ does not depend on the normal choice.

Our earlier computations now give the curvatures directly. With outward normals, the sphere has $H = -1/R, K = 1/R^2$, and the circular cylinder has $H = -1/(2R), K = 0$. For the plane, $H = K = 0$.

For the graph of a C^2 function $h = h(u_1, u_2)$, write

$$f(u_1, u_2) = (u_1, u_2, h(u_1, u_2)).$$

Put $W = \sqrt{1 + h_{u_1}^2 + h_{u_2}^2}$. The upward unit normal is

$$v = \frac{(-h_{u_1}, -h_{u_2}, 1)}{W}.$$

The first fundamental form satisfies

$$g_{ij} = \delta_{ij} + h_{u_i} h_{u_j}, \quad \det g = 1 + h_{u_1}^2 + h_{u_2}^2 = W^2.$$

Since $f_{u_i u_j} = (0, 0, h_{u_i u_j})$, we can read off the second fundamental form:

$$h_{ij} = \langle f_{u_i u_j}, v \rangle = \frac{h_{u_i u_j}}{W}.$$

The symmetric matrix of second partial derivatives is called the **Hessian** of h :

$$\text{Hess } h = D^2 h = \begin{pmatrix} h_{u_1 u_1} & h_{u_1 u_2} \\ h_{u_1 u_2} & h_{u_2 u_2} \end{pmatrix}.$$

Thus $(h_{ij}) = W^{-1} \text{Hess } h$, and

$$K = \frac{\det(h_{ij})}{\det g} = \frac{\det(\text{Hess } h)}{W^4} = \frac{h_{u_1 u_1} h_{u_2 u_2} - h_{u_1 u_2}^2}{(1 + h_{u_1}^2 + h_{u_2}^2)^2}.$$

In particular, the sign of Gaussian curvature is the sign of the Hessian determinant. Reversing the normal changes Π to $-\Pi$, but does not change K .

To understand the shape operator at p , translate and rotate so that $p = f(0, 0) = 0$ and the tangent plane is horizontal. Locally write $f(u_1, u_2) = (u_1, u_2, h(u_1, u_2))$, with

$$h(0, 0) = 0, \quad Dh(0, 0) = 0.$$

Here $f_i = \partial f / \partial u_i$ is a vector, not a component of f . At the origin,

$$f_1(0) = (1, 0, 0), \quad f_2(0) = (0, 1, 0).$$

These vectors are orthonormal, so the first fundamental form has matrix

$$g(0) = (\langle f_i(0), f_j(0) \rangle) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I_2,$$

where I_2 denotes the 2×2 identity matrix.

The upward normal is $\nu(0) = (0, 0, 1)$, while $f_{ij}(0) = (0, 0, h_{u_i u_j}(0))$. Thus the second fundamental form has coefficients

$$h_{ij}(0) = \langle f_{ij}(0), \nu(0) \rangle = h_{u_i u_j}(0).$$

In the coordinate tangent basis, $[L] = g^{-1}(h_{ij})$. Consequently,

$$[L_p] = g(0)^{-1}(h_{ij}(0)) = I_2^{-1} D^2 h(0) = D^2 h(0).$$

The Hessian eigenvalues here are therefore the principal curvatures. This equality uses $Dh(0) = 0$ and the upward normal. For a general graph point, the formula is instead

$$[L] = \frac{1}{\sqrt{1 + |\nabla h|^2}} g^{-1} D^2 h.$$

We can now classify a point using its principal curvatures. For the graph coordinates on the previous page, with $Dh(0) = 0$ and the upward normal, these are the eigenvalues of $D^2h(0)$. Kühnel uses the following terminology:

<i>Point type</i>	<i>Curvature condition</i>	<i>$D^2h(0)$</i>
Elliptic	$K > 0$	Definite
Hyperbolic	$K < 0$	Indefinite
Parabolic	$K = 0, H \neq 0$	Rank one
Level point	$\kappa_1 = \kappa_2 = 0$	Zero
Umbilic	$\kappa_1 = \kappa_2$	λI_2
Properly umbilic	$\kappa_1 = \kappa_2 \neq 0$	$\lambda I_2, \lambda \neq 0$

Here “definite” means positive or negative definite; “indefinite” means that the eigenvalues have opposite signs. The first four types are mutually exclusive and exhaust all points. Umbilicity is an additional property: a level point is umbilic, and a properly umbilic point is elliptic.

Every point of a sphere is properly umbilic, every point of a circular cylinder is parabolic, and every point of a plane is level. A level point need not have a planar neighborhood: for $h = u_1^3 - 3u_1u_2^2$, the origin is a level point.

A **ruled surface** is swept out by a family of straight lines. Locally it has a parametrization

$$f(u, v) = c(u) + v w(u), \quad w(u) \neq 0.$$

For fixed u , varying v traces a line, called a ruling. We work where

$$f_u \times f_v = (\dot{c} + v\dot{w}) \times w \neq 0.$$

Here the dots refer to the arbitrary parameter u .

Since $f_{vv} = 0$, the second fundamental form satisfies $h_{22} = 0$. Therefore every regular ruled surface has

$$K = -\frac{h_{12}^2}{EG - F^2} \leq 0.$$

A ruled surface with $K = 0$ is called **developable**. Thus being ruled alone does not imply $K = 0$.

For example, the cylinder

$$f(u, v) = (R \cos u, R \sin u, 0) + v(0, 0, 1)$$

is ruled and developable. The helicoid

$$f(u, v) = (0, 0, au) + v(\cos u, \sin u, 0), \quad a \neq 0,$$

is ruled but has $K = -a^2/(a^2 + v^2)^2 < 0$.

A **surface of rotation** comes from rotating a plane curve about an axis. For the profile $(r(t), h(t))$ and the z -axis, write

$$f(t, v) = (r(t) \cos v, r(t) \sin v, h(t)).$$

The curves with v fixed are meridians; those with t fixed are parallels. Assume $r > 0$ and $q = \sqrt{\dot{r}^2 + \dot{h}^2} > 0$. Then

$$I = q^2 dt^2 + r^2 dv^2, \quad \nu = \frac{(-\dot{h} \cos v, -\dot{h} \sin v, \dot{r})}{q}.$$

Taking the inner products of the second derivatives with ν gives

$$h_{11} = \frac{\dot{r}\ddot{h} - \ddot{r}\dot{h}}{q}, \quad h_{12} = 0, \quad h_{22} = \frac{r\dot{h}}{q}.$$

Both fundamental forms are diagonal. Thus meridians and parallels are principal directions, with

$$\kappa_1 = \frac{\dot{r}\ddot{h} - \ddot{r}\dot{h}}{q^3}, \quad \kappa_2 = \frac{\dot{h}}{rq}.$$

For example, $r = R, h = t$ gives a cylinder, while $r = a \cosh t, h = at$, with $a > 0$, gives a catenoid.

At $r = 0$ these angular coordinates fail; the surface itself may still be smooth there, as at the poles of a sphere.

A surface is *minimal* when

$$H \equiv 0, \quad \text{equivalently} \quad \kappa_2 = -\kappa_1.$$

Consequently a minimal surface has

$$K = -\kappa_1^2 \leq 0.$$

The two principal curvatures need not vanish: their signed contributions to mean curvature cancel.

For a catenoid, use the rotational formulas with $r(t) = a \cosh t$, $h(t) = at$. They give $q = a \cosh t$ and

$$\kappa_1 = -\frac{1}{a \cosh^2 t}, \quad \kappa_2 = \frac{1}{a \cosh^2 t}.$$

Thus the catenoid is minimal, although it is curved. Planes are minimal too; spheres and circular cylinders are not.

The name comes from the variation of area. On a bounded patch, let $f_\varepsilon = f + \varepsilon \phi \nu$, where ϕ is smooth and vanishes near the boundary. The first variation is

$$\left. \frac{d}{d\varepsilon} \right|_{\varepsilon=0} \text{Area}(f_\varepsilon) = -2 \int H \phi \, dA.$$

Hence $H = 0$ means that area is stationary under every such variation. It does not by itself assert a global area minimum.

A nonzero tangent vector X with $\Pi(X, X) = 0$ defines an ***asymptotic direction***. An ***asymptotic curve*** is a regular curve whose tangent is asymptotic at every point:

$$\Pi(\dot{c}, \dot{c}) = 0.$$

For a unit-speed curve $c = f \circ \gamma$, set $n(s) = \nu(\gamma(s))$. Then $n' = -L(c')$. Differentiating $\langle c', n \rangle = 0$ gives

$$\langle c'', n \rangle = -\langle c', n' \rangle = \Pi(c', c').$$

Thus an asymptotic curve has acceleration tangent to the surface. Its ordinary curvature $\|c''\|$ can still be nonzero.

In an orthonormal principal basis, the equation is

$$\kappa_1 x^2 + \kappa_2 y^2 = 0.$$

If $K < 0$, this describes two distinct tangent lines. If $K > 0$, there are no asymptotic directions. If exactly one principal curvature vanishes, there is one asymptotic line; if both vanish, every direction is asymptotic at that point.

On a minimal surface with $K < 0$, the equation reduces to $x^2 - y^2 = 0$, so the two asymptotic directions are perpendicular. Conversely, perpendicular asymptotic directions force $\kappa_1 + \kappa_2 = 0$.

Every ruling of a ruled surface is an asymptotic curve, since its unit-speed parametrization has $c'' = 0$.

A regular parametrization $f : U \subset \mathbb{R}^2 \rightarrow M$ is **conformal**, or **angle preserving**, if

$$g = \lambda(u, v) \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad \lambda > 0.$$

The coordinates u, v are then called **isothermal parameters**. Equivalently,

$$E = G = \lambda, \quad F = 0, \quad I = \lambda(du^2 + dv^2).$$

The coordinate tangent vectors must be perpendicular *and* have the same length. Orthogonality alone is insufficient.

For parameter vectors $a, b \neq 0$,

$$\langle Df(a), Df(b) \rangle = \lambda \langle a, b \rangle, \quad \|Df(a)\| = \sqrt{\lambda} \|a\|.$$

The factor λ therefore cancels in the cosine formula for the angle. Lengths can change, but every direction at a point is scaled by the same factor $\sqrt{\lambda}$.

For example, $f(u, v) = (2u, v, 0)$ has orthogonal coordinate curves, but $E = 4 \neq 1 = G$. It changes the right angle between $(1, 1)$ and $(1, -1)$, so it is not conformal.

More generally, a local diffeomorphism $\psi : M \rightarrow \tilde{M}$ is conformal when $\psi^* \tilde{g} = \lambda g$, with $\lambda > 0$. Preserving orientation is a separate requirement.

For a surface of rotation, isothermal parameters can be found by changing only the meridian parameter. Starting from

$$I = q(t)^2 dt^2 + r(t)^2 dv^2,$$

define

$$u(t) = \int_{t_0}^t \frac{q(\xi)}{r(\xi)} d\xi.$$

Since $du/dt = q/r > 0$, the inverse function theorem gives a local inverse $t = \psi(u)$, with $dt/du = r/q$. Therefore

$$I = r(\psi(u))^2 (du^2 + dv^2).$$

This proves conformality wherever the rotational coordinates are regular.

For the catenoid, $q = r = a \cosh t$, so its original parameters are already isothermal:

$$I = a^2 \cosh^2 t (dt^2 + dv^2).$$

Another useful example is the Mercator parametrization of the unit sphere away from its poles:

$$f(u, v) = \frac{1}{\cosh u} (\cos v, \sin v, \sinh u).$$

Direct differentiation gives

$$I = \frac{du^2 + dv^2}{\cosh^2 u}.$$

Angular intervals give local charts. The poles correspond to $u \rightarrow \pm\infty$, rather than finite parameter values.

A *Weingarten surface* satisfies a fixed, nontrivial relation between its principal curvatures:

$$\Phi(\kappa_1, \kappa_2) = 0.$$

One also expresses such relations in terms of H and K . The same relation must hold throughout the surface patch; $\Phi \equiv 0$ would impose no condition.

For example, minimal surfaces satisfy

$$\kappa_1 + \kappa_2 = 0,$$

surfaces of constant mean curvature satisfy $\kappa_1 + \kappa_2 = 2H_0$, and developable surfaces satisfy $\kappa_1\kappa_2 = 0$.

The ellipsoid of rotation

$$\frac{x^2 + y^2}{a^2} + \frac{z^2}{c^2} = 1$$

satisfies the less familiar relation

$$\kappa_1 = \frac{a^4}{c^2} \kappa_2,$$

where κ_1 is the meridian curvature and κ_2 the parallel curvature. Here $a, c > 0$ are fixed semiaxes.

For a surface of rotation both curvatures depend only on the meridian parameter. Eliminating that parameter, where possible, produces a Weingarten relation.

A Weingarten *map* exists on every oriented surface patch. Being a Weingarten *surface* is an additional condition on the eigenvalues of that map.